Sebastian Ren

UCL Robotics & AI MEng

a2942159113@gmail.com London, UK

EDUCATION

University College London *Sep 2023 – Jun 2027*
MEng Robotics and Artificial Intelligence, Department of Computer Science *London, UK*

- **GPA:** 3.8/4.0 | **Status:** 4rd Year Undergraduate
- **Relevant Coursework:** Reinforcement Learning, Machine Learning, Robot Dynamics, Modeling & Simulation, Advanced Control Systems, Optimization Filtering & Fusion, Systems Engineering for Real-time Systems

RESEARCH & INTERNSHIP EXPERIENCE

Research Project – UCL Robot Perception & Learning Lab *Nov 2025 – Jan 2026*
Bipedal Locomotion for Quadrupedal Robots with Automatic Fall Recovery *London, UK*
Technologies: PyTorch, PPO, Isaac Gym, Deep Reinforcement Learning, State Estimation

- Reproduced "Learning stable bipedal locomotion skills for quadrupedal robots" (npj Robotics 2025) implementing TumblerNet framework for robust bipedal walking on challenging terrains
- Reproduced Heterogeneous Actor-Critic framework with an RMA-style Estimator, leveraging privileged info for training and inferring real-time states from proprioceptive history.
- Reconstructed complex reward functions based on Variable Height Inverted Pendulum (VHIP) and Cart-Table dynamics to enforce stability during bipedal gaits.
- Bridged the Sim-to-Real gap by implementing an Actuator Net to model non-linear motor dynamics and latency, enabling zero-shot policy deployment on physical hardware.

Research Intern – ANTOBOT *Mar – May 2025*
Human-Following Mobile Robot System *London, UK*
Technologies: YOLOv8n, OSNet Re-ID, Extended Kalman Filter, Artificial Potential Field, ROS, ZED2i RGB-D

- Developed a modular ROS-based system enabling an agricultural robot to detect and follow a specific target in dynamic environments.
- Integrated YOLOv8n for detection and an OSNet-based Re-ID module to distinguish the target from other pedestrians, robust to appearance changes.
- Implemented an Extended Kalman Filter (EKF) for state estimation, enabling continuous tracking during visual occlusions by fusing visual data with odometry.
- Designed an Artificial Potential Field (APF) controller for real-time local path planning and obstacle avoidance while maintaining a safe following distance.
- Validated the system through Gazebo simulations and successful deployment on a physical Antobot platform.

Research Intern – The Future Laboratory, Tsinghua University *Jun – Sep 2024*
Olfaction & Bio-Robotics *Beijing, China*
Technologies: Python, MATLAB, Ansys Fluent, ROS1/2, Raspberry Pi, CAD, 3D Printing

- Conducted research on olfaction and bio-robotics, focusing on insect-inspired airflow sensing mechanisms
- Developed CFD simulations to model airflow around insect antennae during flight using Ansys Fluent
- Designed and implemented data collection programs for electronic-antennae (E-nose) sensor systems
- Led experimental setup design and optimized research methodologies

KEY PROJECTS

Machine Translation of Old Assyrian Cuneiform *Jan 2026*

Built a multi-stage neural machine translation pipeline to translate 4,000-year-old Akkadian cuneiform transliterations into English. Fine-tuned a ByT5-large character-level seq2seq model on 1,500 parallel text pairs, leveraging byte-level tokenization to natively handle rare diacritical characters (š, ṣ, ṭ,) without lossy normalization. Designed a linguistically-informed chunking strategy that splits long documents at Akkadian clause boundaries, improving translation quality on long-form texts. Optimized beam search hyperparameters via Optuna, and implemented vectorized postprocessing for deduplication, fraction normalization, and gap-token standardization.

Vision-Based Non-Contact Anemometer on Embedded Hardware *Nov 2025*

Developed a non-contact anemometer that estimates wind speed by optically tracking the deflection of a gravity-restored drag flap. Built a real-time computer vision pipeline on Raspberry Pi 5 using Python and OpenCV, employing HSV segmentation and centroid extraction to compute deflection angles at 10 Hz. Mapped angles to airspeed via a 3rd-degree polynomial calibration and applied a 1D Kalman filter for noise suppression. Validated in a wind tunnel across 0–10 m/s, achieving RMSE of 0.461 m/s and R^2 of 0.975.

Multi-Modal AI for Mushroom Species & Toxicity Identification *Nov 2025*

Developed a multi-modal AI system integrating YOLOv8n for detection and Vision Transformer (ViT) for fine-grained classification across 16 species, mitigating ViT overfitting through regularization strategies. Fine-tuned Qwen2.5-7B via QLoRA (4-bit) using high-quality synthetic data distilled from a Qwen2.5-VL:32B teacher model, and implemented a RAG system based on ChromaDB to anchor the LLM's expert-level responses in retrieved domain knowledge.

Tree-Climbing Robot with Obstacle Avoidance *Jun 2024*

Collaboratively designed and manufactured a tree-climbing robot from scratch, transforming initial concepts into a fully functional prototype. Developed an electronic control system using Arduino Uno, ultrasonic sensors for branch detection, and motor drivers to actuate custom-designed grippers and a lead screw-based linear lift. Implemented branch avoidance algorithms leveraging ultrasonic sensor data for dynamic decision-making and integrated current sensing mechanisms for motor protection. The robot can climb trees, avoid obstacles, deploy sensors, and ensure safe operation under varying loads.

PUBLICATIONS & AWARDS

Publications

- Toh, B.-Z.†, **Ren, Y.**†, Yu, T. A., Ong, N. K., Mao, Z., & Bano, S. "CasCVS-Net: A Staged Multi-Task Cascade for Critical View of Safety Assessment." *(†Equal contribution)* **MICCAI SafeSurg Workshop (Oral).**
- Pan, Y., **Ren, Y.**, Chan, C., Sun, J., & Zhou, C. "Calf-Integrated Arms for Bimanual Quadruped Loco-Manipulation." **CLAWAR 2026 (Oral).**
- Xu, Y., Zhu, F., Li, Y., **Ren, S.**, Huang, X., & Chen, Y. "RGBSQGrasp: Inferring Local Superquadric Primitives from Single RGB Image for Graspability-Aware Bin Picking." **IROS 2025 RGMCW (Best Workshop Paper Award).**

Awards & Honors

- **Kaggle 18th Place (Silver Medal)**, Deep Past Challenge : Akkadian–English Translation 2026
- **Gold Medal**, UKMT Mathematical Challenge 2023
- **Gold Medal**, British Physics Olympiad 2021